**Q1 - 2024 (01 Feb Shift 2)**

Let $ABC$ be an isosceles triangle in which $A$ is at $(-1,0)$, $\angle \mathrm{A}=\frac{2\pi}{3}$, $\mathrm{AB}=\mathrm{AC}$ and B is on the positive x-axis. If $\mathrm{BC}=4\sqrt{3}$ and the line BC intersects the line $y=x+3$ at $(\alpha,\beta)$, then $\frac{\beta^4}{\alpha^2}$ is :

**Q2 - 2024 (27 Jan Shift 1)**

The portion of the line $4x+5y=20$ in the first quadrant is trisected by the lines $\mathrm{L}_1$ and $\mathrm{L}_2$ passing through the origin. The tangent of an angle between the lines $\mathrm{L}_1$ and $\mathrm{L}_2$ is :

(1) $\frac{8}{5}$

(2) $\frac{25}{41}$

(3) $\frac{2}{5}$

(4) $\frac{30}{41}$

**Q3 - 2024 (27 Jan Shift 2)**

Let R be the interior region between the lines $3x-y+1=0$ and $x+2y-5=0$ containing the origin. The set of all values of $a$, for which the points $\left(a^2, a+1\right)$ lie in $R$, is :

(1) $(-3,-1)\cup\left(-\frac{1}{3},1\right)$

(2) $(-3,0)\cup\left(\frac{1}{3},1\right)$

(3) $(-3,0)\cup\left(\frac{2}{3},1\right)$

(4) $(-3,-1)\cup\left(\frac{1}{3},1\right)$

**Q4 - 2024 (27 Jan Shift 2)**

Let A and B be two finite sets with m and n elements respectively. The total number of subsets of the set A is 56 more than the total number of subsets of $B$. Then the distance of the point $\mathrm{P}(\mathrm{m},\mathrm{n})$ from the point $\mathrm{Q}(-2,-3)$ is

(1) 10

(2) 6

(3) 4

(4) 8

**Q5 - 2024 (27 Jan Shift 2)**

If the sum of squares of all real values of $\alpha$, for which the lines $2x - y + 3 = 0, 6x + 3y + 1 = 0$ and $\alpha x + 2y - 2 = 0$ do not form a triangle is $p$, then the greatest integer less than or equal to $p$ is

**Q6 - 2024 (29 Jan Shift 1)**

In a $\triangle ABC$, suppose $y = x$ is the equation of the bisector of the angle B and the equation of the side $AC$ is $2x - y = 2$. If $2AB = BC$ and the point $A$ and B are respectively $(4, 6)$ and $(\alpha, \beta)$, then $\alpha + 2\beta$ is equal to

(1) 42

(2) 39

(3) 48

(4) 45

**Q7 - 2024 (29 Jan Shift 1)**

Let $\left(5, \frac{a}{4}\right)$, be the circumcenter of a triangle with vertices $A(a, -2), B(a, 6)$ and $C\left(\frac{a}{4}, -2\right)$. Let $\alpha$ denote the circumradius, $\beta$ denote the area and $\gamma$ denote the perimeter of the triangle. Then $\alpha + \beta + \gamma$ is

(1) 60

(2) 53

(3) 62

(4) 30

**Q8 - 2024 (29 Jan Shift 2)**

The distance of the point $(2, 3)$ from the line $2x - 3y + 28 = 0$, measured parallel to the line $\sqrt{3}x - y + 1 = 0$, is equal to

(1) $4\sqrt{2}$

(2) $6\sqrt{3}$

(3) $3+4\sqrt{2}$

(4) $4+6\sqrt{3}$

**Q9 - 2024 (29 Jan Shift 2)**

Let A be the point of intersection of the lines 3x + 2y = 14, 5x – y = 6 and B be the point of intersection of the lines 4x + 3y = 8, 6x + y = 5. The distance of the point P(5, –2) from the line AB is

(1) $\frac{13}{2}$

(2) 8

(3) $\frac{5}{2}$

(4) 6

**Q10 - 2024 (30 Jan Shift 1)**

A line passing through the point $\mathrm{A}(9,0)$ makes an angle of $30^{\circ}$ with the positive direction of $x$-axis. If this line is rotated about A through an angle of $15^{\circ}$ in the clockwise direction, then its equation in the new position is

(1) $\frac{y}{\sqrt{3}-2}+x=9$

(2) $\frac{x}{\sqrt{3}-2}+y=9$

(3) $\frac{x}{\sqrt{3}+2}+y=9$

(4) $\frac{y}{\sqrt{3}+2}+x=9$

**Q11 - 2024 (30 Jan Shift 2)**

If $x^2-y^2+2hxy+2gx+2fy+c=0$ is the locus of a point, which moves such that it is always equidistant from the lines x + 2y + 7 = 0 and 2x − y +8 = 0, then the value of g + c + h − f equals

(1) 14

(2) 6

(3) 8

(4) 29

**Q12 - 2024 (31 Jan Shift 1)**

Let $\alpha, \beta, \gamma, \delta \in \mathrm{Z}$ and let $\mathrm{A}(\alpha, \beta), \mathrm{B}(1,0), \mathrm{C}(\gamma, \delta)$ and $D(1,2)$ be the vertices of a parallelogram $ABCD$. If $AB=\sqrt{10}$ and the points $A$ and $C$ lie on the line $3y=2x+1$, then $2(\alpha+\beta+\gamma+\delta)$ is equal to

(1) 10

(2) 5

(3) 12

(4) 8

**Q13 - 2024 (31 Jan Shift 2)**

Let $A(a,b), B(3,4)$ and $(-6,-8)$ respectively denote the centroid, circumcentre and orthocentre of a triangle. Then, the distance of the point $P(2a+3, 7b+5)$ from the line $2x+3y-4=0$ measured parallel to the line $x-2y-1=0$ is

(1) $\frac{15\sqrt{5}}{7}$

(2) $\frac{17\sqrt{5}}{6}$

(3) $\frac{17\sqrt{5}}{7}$

(4) $\frac{\sqrt{5}}{17}$

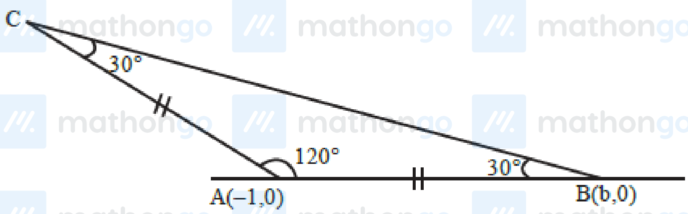
**Q14 - 2024 (31 Jan Shift 2)**

Let $A(-2,-1), B(1,0), C(\alpha, \beta)$ and $D(\gamma, \delta)$ be the vertices of a parallelogram $ABCD$. If the point $C$ lies on $2x-y=5$ and the point $D$ lies on $3x-2y=6$, then the value of $|\alpha+\beta+\gamma+\delta|$ is equal to _______.

# Answer Key

| | | | |
|---|---|---|---|
| **Q1** (36) | **Q2** (4) | **Q3** (2) | **Q4** (1) |
| **Q5** (32) | **Q6** (1) | **Q7** (2) | **Q8** (4) |
| **Q9** (4) | **Q10** (1) | **Q11** (1) | **Q12** (4) |
| **Q13** (3) | **Q14** (32) | | |

**Q1**




$\frac{c}{\sin 30^\circ} = \frac{4\sqrt{3}}{\sin 120^\circ}$ [By sine rule]

$2c = 8 \Rightarrow c = 4$

$AB = |(b+1)| = 4$

$b = 3,\ m_{AB} = 0$

$m_{BC} = \frac{-1}{\sqrt{3}}$

$BC : -y = \frac{-1}{\sqrt{3}}(x-3)$

$\sqrt{3}y + x = 3$

Point of intersection : $y = x + 3, \sqrt{3}y + x = 3$

$(\sqrt{3}+1)y = 6$

$y = \frac{6}{\sqrt{3}+1}$

$x = \frac{6}{\sqrt{3}+1} - 3$

$= \frac{6 - 3\sqrt{3} - 3}{\sqrt{3}+1}$

$= 3\frac{(1-\sqrt{3})}{(1+\sqrt{3})} = \frac{-6}{(1+\sqrt{3})^2}$

$\frac{\beta^4}{\alpha^2} = 36$

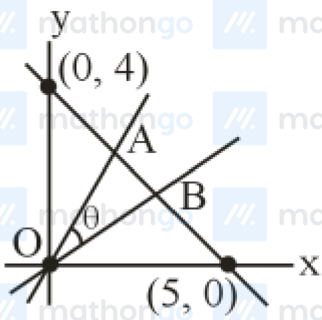
**Q2**

Co-ordinates of A $= \left(\frac{5}{3}, \frac{8}{3}\right)$

Co-ordinates of B $= \left(\frac{10}{3}, \frac{4}{3}\right)$

Slope of OA $= m_1 = \frac{8}{5}$

Slope of OB $= m_2 = \frac{2}{5}$




$$\tan\theta=\left|\frac{m_1-m_2}{1+m_1m_2}\right|$$

$$\tan\theta=\frac{\frac{6}{5}}{1+\frac{16}{25}}=\frac{30}{41}$$

$$\tan\theta=\frac{30}{41}$$

**Q3**

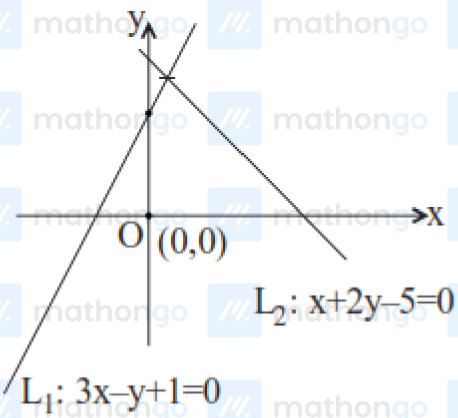
$P\left(a^2, a+1\right)$

$L_1=3x-y+1=0$

Origin and P lies same side w.r.t. $L_1$

$\Rightarrow L_1(0)\cdot L_1(P)>0$

$\therefore 3\left(a^2\right)-(a+1)+1>0$




$\Rightarrow 3a^2-a>0$

$a\in(-\infty,0)\cup\left(\frac{1}{3},\infty\right)\ldots(1)$

Let $L_2: x+2y-5=0$

Origin and P lies same side w.r.t. $L_2$

$\Rightarrow L_2(0) \cdot L_2(P) > 0$

$\Rightarrow a^2 + 2(a+1) - 5 < 0$

$\Rightarrow a^2 + 2a - 3 < 0$

$\Rightarrow (a+3)(a-1) < 0$

$\therefore a \in (-3, 1) \ldots\ldots\ldots\ldots\ldots (2)$

Intersection of (1) and (2)

$a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$

**Q4**

$2^m - 2^n = 56$

$2^n\left(2^{m-n} - 1\right) = 2^3 \times 7$

$2^n = 2^3$ and $2^{m-n} - 1 = 7$

$\Rightarrow n = 3$ and $2^{m-n} = 8$

$\Rightarrow n = 3$ and $m - n = 3$

$\Rightarrow n = 3$ and $m = 6$

$P(6, 3)$ and $Q(-2, -3)$

$PQ = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$

Hence option (1) is correct

**Q5**

$2x - y + 3 = 0$

$6x + 3y + 1 = 0$

$\alpha x + 2y - 2 = 0$

Will not form a $\Delta$ if $\alpha x + 2y - 2 = 0$ is concurrent with $2x - y + 3 = 0$ and $6x + 3y + 1 = 0$ or parallel to either of them so

Case-1: Concurrent lines

$$\begin{vmatrix} 2 & -1 & 3 \\ 6 & 3 & 1 \\ \alpha & 2 & -2 \end{vmatrix} = 0 \Rightarrow \alpha = \frac{4}{5}$$

Case-2 : Parallel lines

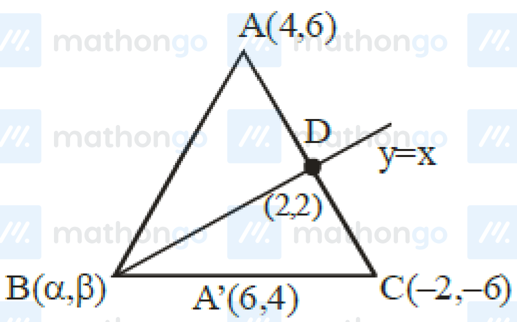
$-\frac{\alpha}{2} = \frac{-6}{3}$ or $-\frac{\alpha}{2} = 2$

$\Rightarrow \alpha = 4$ or $\alpha = -4$

$P = 16 + 16 + \frac{16}{25}$

$[P] = \left[32 + \frac{16}{25}\right] = 32$

**Q6**




AD : DC = 1 : 2

$\frac{4-\alpha}{6-\alpha}=\frac{10}{8}$

$\alpha=\beta$

$\alpha=14$ and $\beta=14$

**Q7**

$A(a,-2), B(a,6), C\left(\frac{a}{4},-2\right), O\left(5,\frac{a}{4}\right)$

$AO=BO$

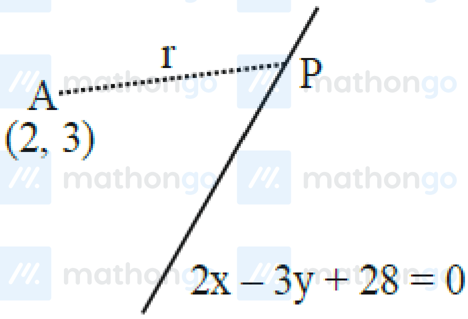
$(a-5)^2+\left(\frac{a}{4}+2\right)^2=(a-5)^2+\left(\frac{a}{4}-6\right)^2$

$a=8$

$AB=8, AC=6, BC=10$

$\alpha=5, \beta=24, \gamma=24$

**Q8**




Writing P in terms of parametric co-ordinates 2 + r

$\cos\theta, 3 + r\sin\theta$ as $\tan\theta = \sqrt{3}$

$P\left(2+\frac{r}{2}, 3+\frac{\sqrt{3}r}{2}\right)$

$P$ must satisfy $2x - 3y + 28 = 0$

So, $2\left(2+\frac{r}{2}\right) - 3\left(3+\frac{\sqrt{3}r}{2}\right) + 28 = 0$

We find $r = 4 + 6\sqrt{3}$

**Q9**

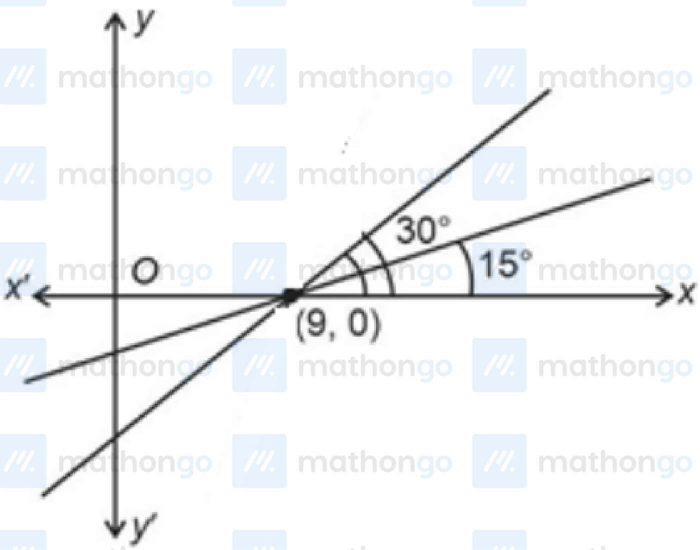
Solving lines $L_1(3x + 2y = 14)$ and $L_2(5x - y = 6)$ to get $A(2, 4)$ and solving lines $L_3(4x + 3y = 8)$ and $L_4(6x + y = 5)$ to get $B\left(\frac{1}{2}, 2\right)$.

Finding Eqn. of AB : $4x - 3y + 4 = 0$

Calculate distance PM

$\Rightarrow \left|\frac{4(5)-3(-2)+4}{5}\right| = 6$

**Q10**




$\mathrm{Eq}^{\mathrm{n}} : y - 0 = \tan 15^\circ(x - 9) \Rightarrow y = (2 - \sqrt{3})(x - 9)$

**Q11**

Cocus of point $\mathrm{P}(\mathrm{x}, \mathrm{y})$ whose distance from Gives $\mathrm{X}+2 \mathrm{y}+7=0 \& 2 \mathrm{x}-\mathrm{y}+8=0$ are equal is

$\frac{x+2y+7}{\sqrt{5}}=\pm\frac{2x-y+8}{\sqrt{5}}\ (x+2y+7)^2-(2x-y+8)^2=0$

Combined equation of lines

$(x-3y+1)(3x+y+15)=0$

$3x^2-3y^2-8xy+18x-44y+15=0$

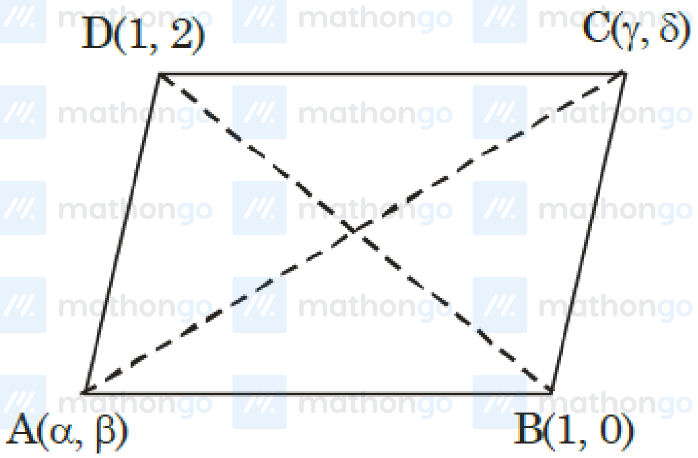
$x^2-y^2-\frac{8}{3}xy+6x-\frac{44}{3}y+5=0$

$x^2-y^2+2hxy+2gx2+2fy+c=0$

$h=\frac{4}{3}, g=3, f=-\frac{22}{3}, c=5$

$g+c+h-f=3+5-\frac{4}{3}+\frac{22}{3}=8+6=14$

**Q12**




Let E is mid point of diagonals

$\frac{\alpha+\gamma}{2}=\frac{1+1}{2}$ $\quad\&\frac{\beta+\delta}{2}=\frac{2+0}{2}$

$\alpha+\gamma=2$ $\quad\beta+\delta=2$

$2(\alpha+\beta+\gamma+\delta)=2(2+2)=8$

**Q13**

$\mathrm{A}(\mathrm{a}, \mathrm{b}), \quad \mathrm{B}(3,4), \quad \mathrm{C}(-6,-8)$

$\mathrm{C}\ \frac{2:1}{A}$

$(-6,-8) \quad (a, b) \quad (3,4)$

$\Rightarrow a=0, b=0 \quad \Rightarrow P(3,5)$

Distance from $P$ measured along $x-2y-1=0$

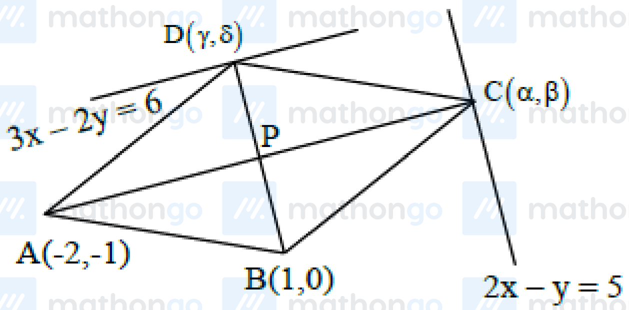
$\Rightarrow x = 3 + r\cos\theta, \quad y = 5 + r\sin\theta$

Where $\tan\theta = \frac{1}{2}$

$r(2\cos\theta + 3\sin\theta) = -17$

$\Rightarrow r = \left|\frac{-17\sqrt{5}}{7}\right| = \frac{17\sqrt{5}}{7}$

**Q14**




$\mathrm{P} \equiv \left(\frac{\alpha-2}{2}, \frac{\beta-1}{2}\right) \equiv \left(\frac{\gamma+1}{2}, \frac{\delta}{2}\right)$

$\frac{\alpha-2}{2} = \frac{\gamma+1}{2}$ and $\frac{\beta-1}{2} = \frac{\delta}{2}$

$\Rightarrow \alpha - \gamma = 3 \ldots. (1), \quad \beta - \delta = 1 \ldots\ldots (2)$

Also, $(\gamma, \delta)$ lies on $3x - 2y = 6$

$3\gamma - 2\delta = 6 \ldots\ldots (3)$

and $(\alpha, \beta)$ lies on $2x - y = 5$

$\Rightarrow 2\alpha - \beta = 5. \ \ldots\ldots (4)$

Solving (1), (2), (3), (4)

$\alpha = -3, \beta = -11, \gamma = -6, \delta = -12$

$|\alpha + \beta + \gamma + \delta| = 32$